TMOS Performance/Tuning

Dan Holland

Agenda

Optimizing TCP/HTTP

Data Collection

Diagnosis

Applying solutions

 © 2026 F5

Our Team

Dan Holland

Principal Solutions Engineer
d.holland@f5.com

Jim Hart

Principal Solutions Engineer
j.hart@f5.com

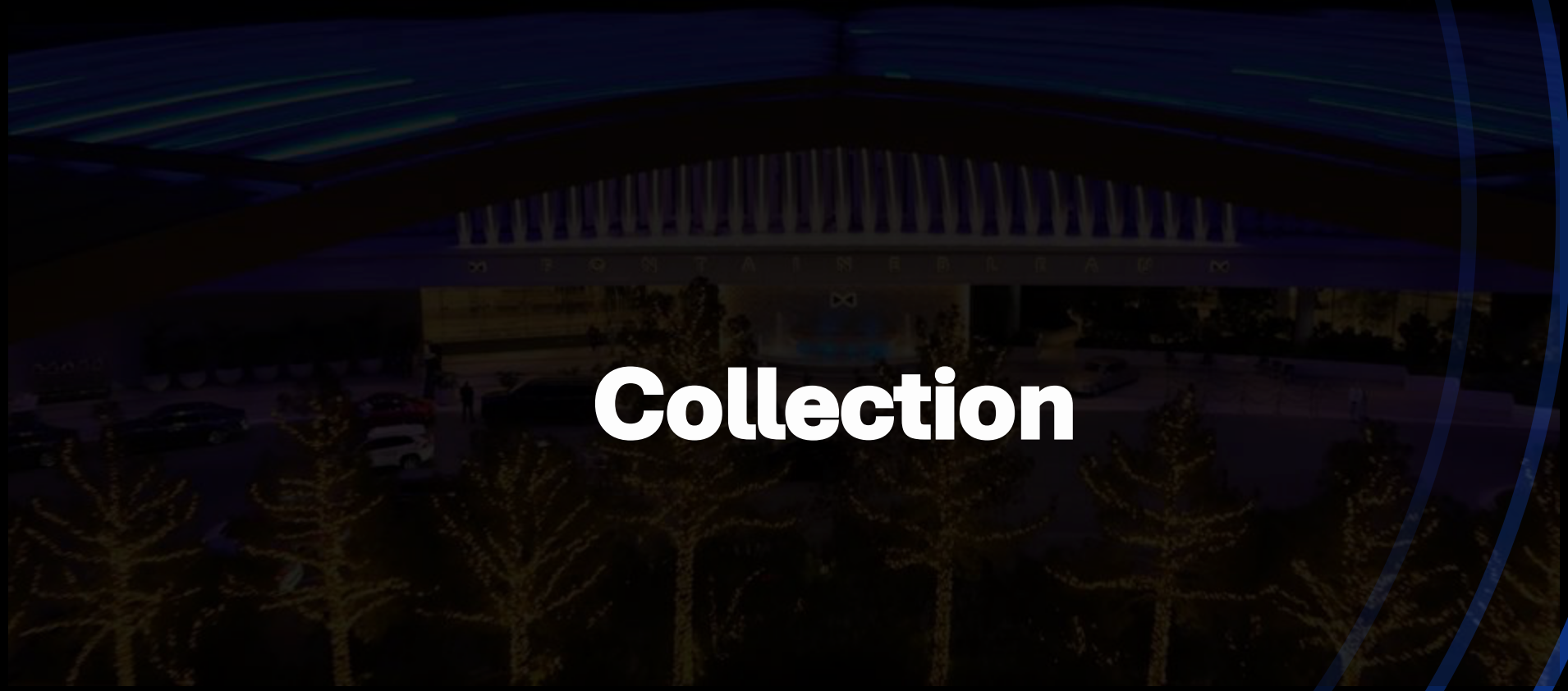
Michael Wang

Sr. Principal Solutions Engineer
michael.wang@f5.com

David Larsen

Manager, Solutions Engineering
David.Larsen@f5.com

 © 2026 F5



Collection

© 2026 F5

Collect: Built-in Analytics & Telemetry

Step 1

AVR	View Statistics	Telemetry Streaming
Create/attach Analytics profile	Analytics for per-Virtual Server	LTM request-log
HTTP/TCP metrics, latency	Per-URL	Syslog off-box
Fall-back on iRule logging	Per-Client	

 © 2026 F5

Collect: Logs & Traces

Step 1

- Request Logging
  - Use HSL profile for HTTP request/response fields
  - Expensive – refine for specific elements, not the whole box
- Tcpdump
  - Specify VLAN or 0.0 for everything
  - Disable DNS (-n) and write to file
  - K411: Overview of packet tracing with the tcpdump utility
- Temporary LB decision logging via iRule (LB::enable_decisionlog)
  - Only for deep selection traces
  - K53385806: Using the LB::enable_decisionlog iRules command

 © 2026 F5

Collect: Custom iRules Logging

Step 1

- Use debug flags or sampling to prevent log floods
- HSL via iRule for non-HTTP workflows
- Use sparingly and tactically – ENSURE YOU TURN OFF after measurement

```
when HTTP_REQUEST {
  set reqURI [HTTP::uri]
}

when HTTP_RESPONSE {
  set resCode [HTTP::status]
  if { $resCode starts_with 4 } {
     log local0. "Response Code: $resCode - URI: $reqURI"
  }
}
```

```
when HTTP_REQUEST {
  set reqURI [HTTP::uri]
}

when HTTP_RESPONSE {
  log local0. "Content-Length (Bytes): [HTTP::header value Content-Length] - URI: $reqURI"
}
```

 © 2026 F5

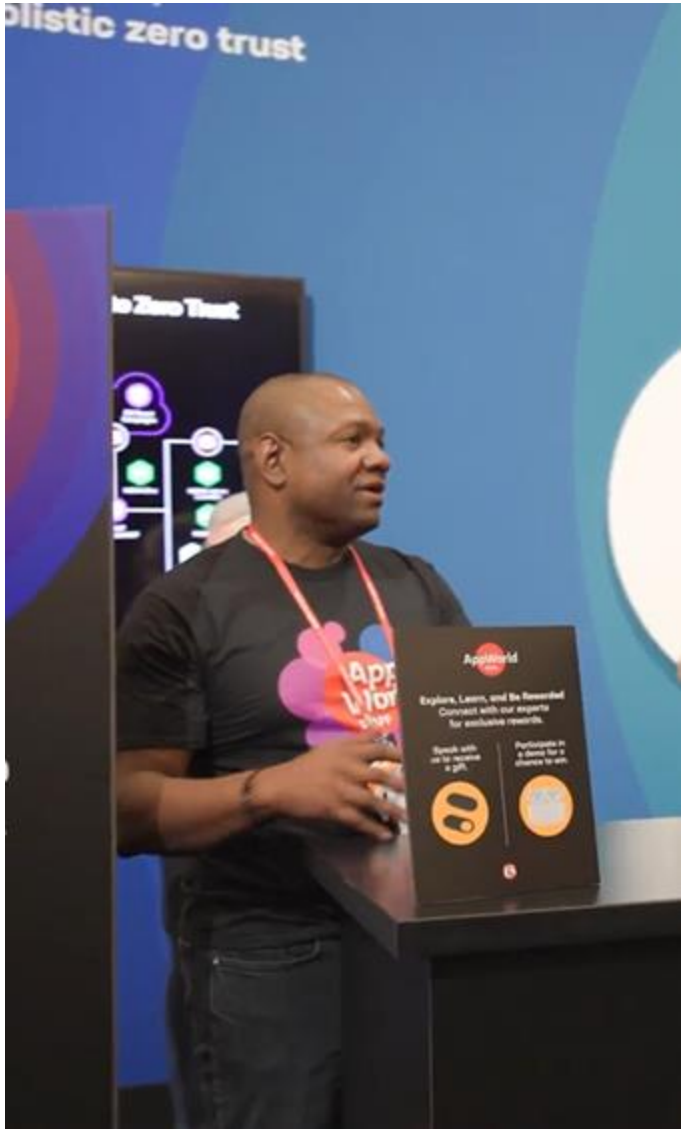
Diagnose

 © 2026 F5

Diagnose: Classic TCP Symptoms & Where to Look

Step 2

- Which timeout is ticking? Map to TCP state diagram (Close-Wait, FIN-Wait, Idle)
- SYN/RTO behavior and Congestion Metrics Cache (CMC)
- Zero-Window / Window-Full patterns indicate buffering/spooling needs
- RST issues
  - K9812: Overview of BIG-IP TCP RST behavior
  - K13223: Configuring the BIG-IP system to log TCP RST packets



 © 2026 F5

Apply

 © 2026 F5

Apply: TCP Profiles and Tuning

Step 3

Pick the Right TCP Profile First

- Start with f5-tcp-progressive
- Leverage Auto Proxy Buffer / Auto Send / Auto Receive

Fine-Tune TCP

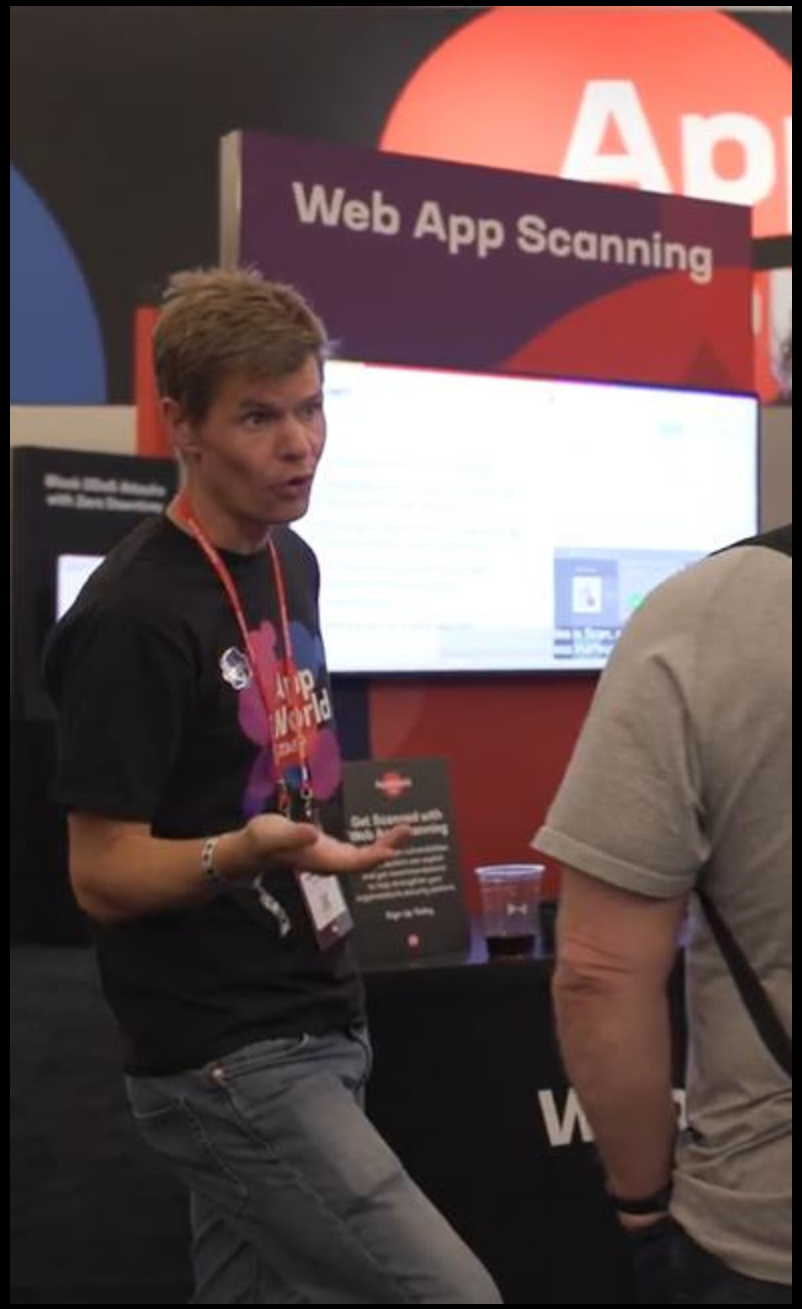
- Minimum RTO sized to RTT + delayed-ACK budget
- Congestion Control
  - Congestion-Control-Algorithms
  - Init cwnd/rwnd (Congestion Window / Receive Window)
- CMC (Congestion Metrics Cache) timeout affects convergence for returning clients

 © 2026 F5

Apply: HTTP Layer Solutions

Step 3

- OneConnect for server-side connection reuse; choose a sane source mask
  - Goal is to reduce repetitive 3-way handshakes, connection initiation time, LB->server congestion
  - LB Resource savings
  - Additional savings with LB->Server TLS connections
- Caching (Web Acceleration) for static assets; understand memory limits
- Selective compression; measure CPU impact



 © 2026 F5

Review

Collect	Diagnose	Apply
AVR/Analytics	Timeouts & SYN/RTO	Start with f5-tcp-progressive + HTTP
Request Logging (HSL)	Zero-Window	Add OneConnect
Tcpdump	OneConnect/HTTP-layer constraints	Evaluate caching/compression
Telemetry		Iterate with data

 © 2026 F5

App World